

Professional Education 8: Assessment and Evaluation of Learning

Basic Concepts

- **Test** - an instrument designed to measure any characteristic, quality, ability, skill or knowledge
- **Measurement** - a process of quantifying the degree to which someone or something possesses a given trait (i.e. quality, characteristics, feature)
- **Assessment** - a process of gathering and organizing quantitative or qualitative data into an interpretable form to have a basis for judgment or decision-making
- **Evaluation** - a process of systematic collection and analysis of both qualitative and quantitative data in order to make some judgment or decision; involves judgment about the desirability of changes in students

Shift of Educational Focus from Content to Learning Outcomes

The Motivation for OBE is... A call for a QUALITY and ACCOUNTABILITY in EDUCATION

OUTCOMES- BASED EDUCATION (OBE) and ASSESSMENT

- Education based on outcome. (**Immediate or deferred outcome**)

Immediate Outcome

- Are competencies/ skills upon completion of a lesson, a subject, a grade/ year, a course or a program itself. Examples are ability to communicate in writing, reading, speaking and solve mathematical problems.

Deferred Outcome

- Refer to the ability to apply cognitive, psychomotor and affective skills/ competencies in the various aspects of the professional and workplace practice. Examples are success in professional practice or occupation as evidence of skill in career planning, health and wellness and continuing education.

OBE, Spady's Version

- Spady Spouses **transformational OBE**. Transformational OBE is concerned with long- term, cross- curricular outcomes that are related directly to student's future life roles such as being a productive worker or a responsible citizen or a parent.
- For Spady, the outcomes he refers to are the deferred outcomes.

Outcome- Based Teaching and Learning (OBTL), Biggs Version

Biggs and Tang (2007) make use of the term outcome- based teaching- learning (OBTL) which in **essence is OBE applied in the teaching- leaning process.**

Learning Outcomes- outcomes which are specific than institutional outcomes, program outcomes and course outcomes.

- Statements that teachers need to expect among students to demonstrate after they have taught.

Outcomes in Different Levels

Institutional Outcomes- graduate attributes that graduates of the institution are expected to demonstrate after graduation.

Program Outcomes – outcomes that graduate of the program are expected to demonstrate at the end of the program.

Course Outcomes – are the particular subject outcomes

Learning Outcomes- are the most specific outcomes that the teacher is concerned with in his/ her specific lessons.

Principles of OBE

1. **Clarity of focus-** outcomes which students are expected to demonstrate at the end of the program are **clear**.
2. **Designing Down-** basing the details of your instructional design on the outcomes, the focus of instruction.
3. **High Expectations** – not all learners can learn the same thing in the same way and in the same amount of time but all are capable of mastery and meaningful learning.
4. **Expanded Opportunities** – Most learners can achieve high standards if they are given appropriate opportunities.

Constructive Alignment

- The assessment tasks and the specific criteria as bases of judgment of student's performance are aligned to the intended learning outcomes.

Program Outcomes for Teacher Education

- a. Articulate the rootedness of education in philosophical, socio-cultural, historical, psychological, and political contexts.
- b. Demonstrate mastery of subject matter/discipline.
- c. Develop innovative curricula, instructional plans, teaching approaches, and resources for diverse learners.
- d. Facilitate learning using a wide range of teaching

Learning Outcomes: Sources and Characteristics

- Learner- centered;
- Aligned with the institutional, program and course outcomes;
- Based on local, national and international trends;
- Very well understood by both learner and students;
- Include a spectrum of thinking (Bloom's Taxonomy of Educational Objectives)

6 Levels in the Cognitive Domain (Bloom's Taxonomy)

- **Remembering**- To recall, recognise, or identify concepts
- **Understanding**- To comprehend meaning
- **Applying**- Use or apply knowledge
- **Analyzing**- Interpret elements
- **Evaluating** - Assess effectiveness of whole concepts
- **Creating**- develop new concepts or approaches

5 Levels in the Affective Domain (Bloom's Taxonomy)

- **Receiving**- Open to experience or idea, willing to hear
- **Responding**- Get involved in or participate actively
- **Valuing**- Attach values and express personal opinions
- **Organizing**- Develop value system
- **Internalising** - Adopt belief system or philosophy

5 Levels in the Affective Psychomotor Domain (Bloom's Taxonomy)

- **Imitation**- To observe and replicate
- **Manipulation**- Reproduce activity from memory
- **Precision**- Demonstrate activity independently
- **Articulation**- Integrate and adapt expertise
- **Naturalisation**- Automated, unconscious mastery activity

Marzano's New Taxonomy

• Marzano's model of thinking skills incorporates a wider range of factors that affect how learners think and provides a research- based theory to help teachers improve their learners' thinking.

3 systems : **Cognitive System**, **Metacognitive System** and **Self- System**

Cognitive System – processes all the necessary information and the knowledge which consists of information, mental procedures and physical procedures.

Metacognitive System – sets goal and keeps track of how well these goals are being achieved

Self- System – decides whether to continue the current behavior or engage in a new learning activity.

Determining the Progress Towards the Attainment of Learning Outcomes

Purposes of Assessment

- Assessment **for** learning (Ex. placement, formative, diagnostic)
- Assessment **of** learning (Ex. summative)
- Assessment **as** learning (training teachers on how to assess)

Assessment

Traditional Assessment

- refers to pen and paper mode of assessing any quality, ability, skill or knowledge (Ex. standardized and teacher-made tests) When teacher makes use of traditional assessment, he/ she uses pen- and- paper test.
- **Pen- and Paper Test** can be in the form of **selected- response test** or **constructed- response test**

- When a teacher makes use of a non- traditional or authentic assessment, he/ she uses non- paper-and pencil test like a performance test which is done through direct demonstration of a skill in a process or procedure or through a presentation of a product.

Alternative Assessment

- **Performance-based Assessment** - a mode of assessment that requires the students to perform a significant task that is relevant to a task outside the school (Ex. practical test, oral and aural tests, projects)
- **Portfolio Assessment** - a process of gathering multiple indicators of student progress to support course goals in dynamic, ongoing and collaborative process

Diagnostic assessment

- is pre- instruction assessment.
- Used to determine learner’s misconceptions to guide the teacher on how to go about instruction.
- Done through a pre- test.

Formative assessment

- is assessment during instruction.
- Used to find out if students can follow the lesson or not.
- Teacher adjusts instruction to ensure that students learn the basic knowledge and skills to attain the intended learning outcome.

Summative Assessment

- Assessment at the end of the instruction.
- It is the basis of grading.

Norm- referenced assessment

- Compares performance of learners with the performance of other learners

Criterion- referenced Assessment

- Compares learners’ performance to established criteria or standard of success.

Contextualized Assessment

- Focuses on student’s application of knowledge, skills and values in the context of work related to specialization

Decontextualized Assessment

- Focuses on student’s knowledge and skills which are not necessarily connected to work context.

Authentic Assessment

- refers to the use of assessment methods that simulate true-to-life situations.
- a form of assessment in which students are asked to perform real- world tasks that demonstrate meaningful application of essential knowledge and skills

Attributes

1. Action
2. Setting
3. Method
4. Focus
5. Outcome

Traditional Assessment

- Selecting a response
- Contrived
- Recall
- Teacher- structured
- Indirect evidence

Authentic assessment

- Performing a task
- Simulation
- Application
- Student- Structured
- Direct evidence

Non- Test assessment of Learning

- An alternative assessment in the sense that it diverts from the paper- and- pen test
- Formative assessment which is an ongoing process to give feedback to students to increase their competence.

Examples of Non- Test assessment

Portfolio – a purposeful collection of student’s works that exhibit the student’s efforts, progress, and achievements in one or more areas of the curriculum.

Teacher Observation- The teacher observes the students while they work to make certain the students understand the assignment and are on task.

Example: Cooperative Learning

Slates or Hand Signals- means of signaling answers to the teacher

Example: Review Questions _ write answers and hold up slate

Daily Assignments – The student completes the work assigned on a daily basis to be completed at school or home.

Example: Worksheet or research

Journal- Student's write daily on assigned or personal topics.

Example: What do you remember in yesterday's lesson.

Games – Teachers utilize fun activities to have students practice and review concepts.

Example: Science Trivia

Projects- After students are taught the basics of triangles, such as types, congruence, proofs and similarity, each student is asked to create a poster, drawing, or personal adornment such as fingernail design, piece of jewelry, or tattoo that uses at least two different triangular shapes.

Debates- The students taking opposing positions on a topic and defend their position.

Example: The pros and cons of environmental legislations.

Checklist- The teacher will make a list of objectives that students need to master and then check off the skill as the student masters it.

Cartooning- Students will use drawings to depict situation and ideas. *Example: environmental issues*

Models – The students produce a miniature replica of a given topic. *Example: planetarium*

Notes- Students will write a summary of the lesson. *Example: Outline of the day's lesson*

Panel Discussion- A group of students verbally present information.

Example: A discussion presenting both pros and cons of the environmental issues.

Demonstration - Students present a visual enactment of a particular skill or activity.

Example: proving that air has weight

Transversal Competencies

- are competencies that are transferable between jobs. People use to call them experience, soft skills, emotional intelligence and employability skills.

Principles of High Quality Assessment

- | | |
|---|--|
| A. Clear and appropriate learning targets | G. Practicality and efficiency |
| B. Appropriate methods | H. Assessment should be a continuous process |
| C. Balance | I. Authenticity |
| D. Validity | J. Communication |
| E. Reliability | K. Positive consequences |
| F. Fairness | L. Ethics |

Assessment is of high quality when:

- It is in accordance with contemporary view or active learning and motivation
- It involves learners in the assessment process beginning with the setting of goals, monitoring of their own learning and in building self- confidence because learners are intrinsically motivated to learn.
- It is not just students' meaningless reproduction of knowledge learned but linking information to other bits of information meaningfully, critically and creatively to apply what they learn to real- world situations.
- It is **valid**. It assesses what is supposed and intended to assess.
- It is **reliable** meaning the test procedures consistent scores.
- It is **fair** like when learners are assessed on something they were taught and is not against subgroups of students.

Some current trends in assessment are:

- Using not only objective tests that measure simple recall but a combination of objective tests and performance-based with emphasis on the latter assessing higher- level cognitive skills.
- Integrated rather than isolated skills using multiple assessment methods from multiple choice test to essay, an interview, a project, a portfolio to self- evaluation.
- Setting high performance and challenging standards including world- class standards for interpreting results.
- involving students in all aspects of assessment from the setting of expected targets to checking on their progress in the course of the teaching- learning process and in determining the extent to which they realized expected targets.
- Making standards and criteria known to all students and even parents rather than be secretive about them to provide a clear picture of a desired performance.
- Using computers as part of assessment

Characteristics of 21st Century Assessment

- | | |
|--------------|--------------------|
| • Responsive | • Multiple Methods |
| • Flexible | • Communicated |

- Integrated
- Informative
- Technically Sound
- Systematic

Performance-based Assessment

- A process of gathering information about student’s learning through actual demonstration of essential and observable skills and creation of products that are grounded in real world contexts and constraints

Types of Performance-based Task

- Demonstration-type - requires no product (Ex. cooking demonstrations, entertaining tourists)
- Creation-type - requires tangible products (Ex. project plan, Research paper, Project flyers *Criteria in Selection a Task*)

GOAL- Provide a statement of the task

- Establish the goal of the task; state the problem, challenge or obstacle in the task.

ROLE- Define the role of the students in the task.

- State the job of the students for the task.

AUDIENCE – Identify the target audience within the context of the scenario.

SITUATION – Explain the situation. What’s the context? What is the challenge?

PRODUCT – Clarify what the students will create and how they will create it.

STANDARDS and CRITERIA – identify the specific standards for success; Give rubric to the students to develop them with the students.

How to assess performance?

- Identify the **competency** that has to be demonstrated by the students with or without a product.
- Describe the **task** to be performed by the students either individually or as a group, the resources needed, time allotment and other requirements to be able to assess the focused competency.

Criteria in Selecting a Performance Task

- | | |
|---------------------|----------------|
| A. Generalizability | E. Feasibility |
| B. Authenticity | F. Scorability |
| C. Multiple foci | G. Fairness |
| D. Teachability | |

- Develop a **scoring rubric** reflecting the criteria, levels of performance and the scores.

Portfolio Assessment

- A purposeful, ongoing, dynamic, and collaborative process of gathering multiple indicators of the learner’s growth and development
- Also performance-based but more authentic than any other performance-based task

Principles of Portfolio Assessment

- Content principle – should reflect important subject matter
- Learning principle – should enable students to become more active learners
- Equity principle – should allow students to demonstrate their learning styles and intelligence

Types of Portfolios

Portfolios come in three types

- **Working portfolio** – a collection of a student’s day-to-day works which reflect his/her learning
- **Show portfolio** – a collection of a student’s best works
- **Documentary portfolio** – a combination of a working and a show portfolio Steps in Portfolio
- **Assessment Portfolio** – is intended to document what a student has learned based on intended learning

outcomes.

Essential Elements of a Portfolio

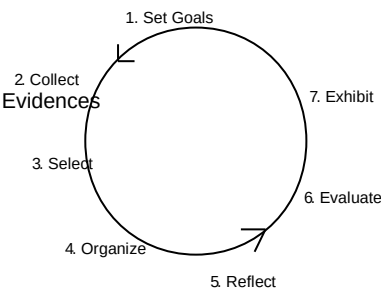
- o Cover letter
- o Table of Contents
- o Entries
- o Dates Entries
- o Drafts

- o Reflections

What a Portfolio includes?

- **Artifacts**– are documents or products that are produced as a result of academic classroom work.
- **Reproductions** – are documentations of a student’s work outside the classroom.
- **Attestations** – are the teacher’s or other responsible person’s documentation to attest to the student progress.
- **Productions** – are the documents that the student himself/ herself prepares.

Steps in Portfolio Development



e- Portfolio

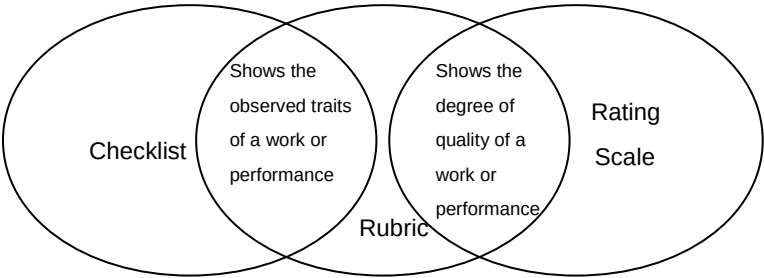
- **Student- Centered e- portfolios** – serve the formative purpose of assessment for learning.
- **School- Centered e- portfolios** – administered by the school for summative assessment purposes.
- **Assessment e- portfolios** – are meant to support assessment of institutional/ school outcomes and are therefore intended for the school as a whole.
- **Learning e- portfolios** – are meant for learners to help them examine and reflect on their own learning.
- **Career/ transfer e- portfolios** – are meant to provide students with a tool for showcasing their achievements to employers of transfer institutions.

A measuring instrument used in rating performance-based tasks
Offers a set of guidelines or descriptions in scoring different levels of performance or qualities of products

Rubrics

- A measuring instrument used in rating performance-based tasks
- Offers a set of guidelines or descriptions in scoring different levels of performance or qualities of products

of learning Similarity of Rubric with Other Scoring Instruments A rubric is a modified checklist and rating scale.



Types of Rubrics

Type	Description
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Holistic Rubric	Describes the overall quality of a performance or product; there is only one rating given to the entire work or performance
Analytic Rubric	Describes the quality of a performance or product in terms of the identified dimensions and/or criteria which are rated independently to give a better picture of the quality of work or performance

Parts of a Scoring Rubric

- Criteria
- Descriptions of the level of performance

Important Elements of a Rubric

Whether holistic or analytic, the rubric should have the following information

- **Competency to be tested** – this should
 - be a behavior that requires either a demonstration or creation of products of learning
- **Performance task** – the task should be authentic, feasible, and has multiple foci
- **Evaluative criteria and their indicators** – these should be made clear using observable traits
- **Performance levels** – these levels could vary in number from 3 or more
- **Qualitative and quantitative descriptions of each performance level** – these descriptions should be observable to be measurable

Tests

Purposes/Uses of Tests

- Instructional (Ex. grouping learners for instruction within a class, identifying learners who need corrective and enrichment experiences, assigning grades)
- Guidance (Ex. preparing information/data to guide conferences with parents about their children, determining interests in types of occupations not previously considered or known by the students)
- Administrative (Ex. determining emphasis to be given to the different learning areas in the curriculum, determining appropriateness of the school curriculum for students of different levels of ability)

Types of Tests

According to:		
What it measures (Purpose)	Educational Test • Aims to measure the <u>results of instruction</u> • Administered after the instructional process Example: Achievement Test	Psychological Test • Aims to measure students intelligence or mental ability in a large degree without reference to what the student has learned • <u>Intangible aspects of an individual</u> • Administered before the instructional process Examples: Aptitude Test, Personality Test, Intelligence Test
How it is interpreted (Interpretation)	Norm-Referenced Test • Result is interpreted by <u>comparing one student with other students</u> • Some will really pass • There is competition for a limited percentage of high score. • Describes student's performance compared to others	Criterion-Referenced Test • Result is interpreted by <u>comparing a student against a set of criteria</u> • All or none may pass • There is NO competition for a limited percentage of high score. • Describes student's mastery of the course objective
The scope of the test (Scope and Content)	Survey • Covers a <u>broad range of objectives</u> • Measures general achievement in certain subjects	Mastery Test • Covers a <u>specific learning objective</u> • Measures fundamental skills and abilities

	Is constructed by trained professional	Is typically constructed by the teacher	
Level of difficulty of the test and time allotment (Time Limit and Level of Difficulty)	Power - Consists of items of <u>increasing level of difficulty but taken with ample time</u> - Measures a student's ability to answer more and more difficult items	Speed - Consists of items with <u>the same level of difficulty but taken with time limit</u> - Measures student's speed and accuracy in responding	
Manner of administration	Individual - Given to <u>one student at a time</u> - Mostly given orally or requires actual demonstration of skill - Many opportunities for clinical observation - Chance to follow-up examinee's response in order to clarify	Group - Given to <u>many individuals at the same time</u> - Usually a pencil and paper test - Lack of insights about the examinee - Same amount of time needed to gather information from each student (i.e. efficient)	
Language mode	Verbal <u>Words are used</u> by students in attaching meaning to or responding to test items	Non-Verbal <u>Pictures or symbols are used</u> by students in attaching meaning to or in responding to test items	
Who constructed the test and who can take it (Construction)	Standardized <u>Made by an expert</u> ; tried out, so it can be used to a wider group - Covers a broad range of content covered in a subject area - Uses mainly multiple choice - Items written are screened and the best items are chosen for the final instrument - Can be scored by a machine - Interpretation of results is usually norm-referenced	Informal <u>Made by the classroom teacher</u> ; not tried out - Covers a narrow range of content - Various types of items are used - Teacher picks or writes items as needed for the test - Scored by a teacher - Interpretation of results is usually criterion-referenced	
Degree of influence of the rater on the outcome (Effect of Biases)	Objective <u>Scorer's personal biases do not affect scoring</u> - Worded so that only one answer satisfies the requirement of the statement - Little or no disagreement on what is the correct answer	Subjective <u>Affected by scorer's personal bias</u>, opinion, or judgment - Several answers are possible - Possible disagreement on what is the correct answer	
Format	Selective Test - There are <u>choices for the answer</u>. - Can be	Supply Test - There are <u>no choices for the answer</u>. - Preparation of	Essay Test Examples: Restricted Response

	answered quickly • Prone to guessing • Time consuming to construct Examples: • Multiple choice • True-False or Alternative Response • Matching Type	items is relatively easy because only a few questions are needed • Lessen the chance of students guessing the correct answer • Bluffing is a problem • Time consuming to score Examples: • Short Answer • Completion Test	• Extended Response
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Assessment of Affective and Other Non-Cognitive Learning Outcomes

Affective/Non-Cognitive Learning Outcome	Sample Behavior
Social attitudes	Concern for the welfare of others, sensitivity to social issues, desire to work toward social improvement
Scientific attitude	Open-mindedness, risk taking and responsibility, resourcefulness, persistence, humility, curiosity
Academic self-concept	Self-perception as a learner in particular subjects (Math, etc.)
Interests	Feeling toward various educational, mechanical, aesthetic, social, recreational, vocational activities
Appreciations	Feelings of satisfaction and enjoyment expressed toward nature, music, art, literature, vocational activities
Adjustments	Relationship to peers, reaction to praise and criticism, emotional, social stability, acceptability

Stages in the Development and Validation of an Assessment Instrument

Phase I: Planning Stage	Phase II: Item Writing Stage	Phase III: Try Out Stage	Phase IV: Evaluation Stage
- Specify the objectives/skills and content area to be measured - Prepare the Table of Specifications - Decide on the item format (Short answer, etc)	- Write test items based on the Table of Specifications - Consult with experts (subject teacher, test expert) for validation (content) and editing	- Conduct First Trial Run (50 to 100 students) - Score - Perform First Item Analysis - Perform First Option Analysis - Revise the test items based on the results of the item analysis - Conduct Second Trial Run/Field Testing - Perform Second Item Analysis - Perform Second Option Analysis - Write the final form of the test The smaller the Difficulty Index is, the more difficult the item. The larger the Discrimination Index is, the more discriminating the item.	- Administration of the final form of the test - Establish test validity - Establish test reliability

Interpreting the Difficulty and Discrimination Indices

Difficulty Index	Interpretation	Discrimination Index	Interpretation
0.00 – 0.20	Very difficult item	-1.00 – -0.60	Questionable item
0.21 – 0.40	Difficult item	-0.59 – -0.20	Not discriminating item
0.41 – 0.60	Moderately difficult item	-0.21 – 0.20	Moderately discriminating item
0.61 – 0.80	Easy item	0.21 – 0.60	Discriminating item
0.81 – above	Very easy item	0.61 – 1.00	Very discriminating item

General Suggestions in Test Writing

1. Use your test specifications as guide to item writing.
2. Write most items than needed.
3. Write the test items well in advance of the testing date.
4. Write each test item so that the task to be performed is clearly defined.
5. Write each test item in appropriate reading level.
6. Write each test item so that it does not provide help in answering other items in the test.
7. Write each test item so that the answer is one that would be agreed upon by the experts.
8. Write each test item so that it is in the proper level of difficulty.
9. Whenever a test is revised, recheck its relevance.

Suggestions for Writing Selective Type Tests

1. Multiple Choice
 - a) The stem of the item should be meaningful by itself and should present a definite problem.
 - b) The item should include as much of the item and should be free of irrelevant material.
 - c) Use a negatively stated item stem only when significant learning outcomes required it.
 - d) Highlight negative words in the stem for emphasis.
 - e) All the alternatives should be grammatically consistent with the stem of the item.
 - f) An item should only have one correct or clearly best answer.
 - g) Items used to measure understanding should contain novelty, but beware of too much. h) All distracters should be plausible.
 - i) Verbal associations between the stem and the correct answer should be avoided.
 - j) The relative length of the alternatives should not provide a clue to the answer.
 - k) The alternatives should be arranged logically.
 - l) The correct answer should appear in each of the alternative positions and approximately equal number of times but in random order.
 - m) Use of special alternatives such as "none of the above" or "all of the above" should be done sparingly.
 - n) Do not use multiple-choice items when other types are more appropriate.
 - o) Always have the stem and alternatives on the same page.
 - p) Break any of these rules when you have a good reason for doing so.
2. Alternative Response
 - a) Avoid broad statements.
 - b) Avoid trivial statements.
 - c) Avoid the use of negative statements, especially double negatives.
 - d) Avoid long and complex sentences.
 - e) Avoid including two ideas in one statement unless cause-effect relationships are being measured.
 - f) If opinion is used, attribute it to some source unless the ability to identify opinion is being specifically measured.
 - g) True statements and false statements should be approximately equal in length.
 - h) The number of true statements and false statements should be approximately equal.
 - i) Start with false statement since it is a common observation that the first statement in this type is always positive.
3. Matching Type
 - a) Use only homogeneous material in a single matching exercise.
 - b) Include an unequal number of responses and premises, and instruct the students that responses may be used once, more than once, or not at all.
 - c) Keep the list of items to be matched brief, and place the shorter responses at the right.
 - d) Arrange the list of responses in logical order.
 - e) Indicate in the directions the basis for matching the responses and premises.
 - f) Place all the items for one matching exercise on the same page.

Suggestions for Writing Supply Type Tests

1. Word the item(s) so that the required answer is both brief and specific.
2. Do not take statements directly from textbooks as a basis for short answer items.
3. A direct question is generally more desirable than an incomplete statement.
4. If the item is to be expressed in numerical units, indicate the type of answer wanted.
5. Blanks for answers should be equal in length.
6. Answers should be written before the item number for easy checking.
7. When completion items are to be used, do not have too many blanks. Blanks should be within or at the end of the sentence and not at the beginning.

Suggestions for Writing Essay Type Tests

1. Restrict the use of essay questions to those learning outcomes that cannot be satisfactorily measured by objective items.

- 2. Avoid the use of optional questions.
- 3. Indicate the approximate time limit or the number of points for each question.
- 4. Prepare the scoring guide (rubric) for the essay questions.

Criteria to Consider when Constructing Good Test Items

Validity

Validity is the degree to which the test measures what it intended to measure. It is the usefulness of the test for a given purpose.

Types of Validity

- **Face Validity** – done by examining the physical appearance of the test
- **Content Validity** – done through a careful and critical examination of the objectives of the test so that it reflects the curricular objectives
- **Criterion-related Validity** – established statistically such that a set of scores revealed by a test is correlated with the scores obtained in another external predictor or measure
 - *Concurrent Validity* – describes the present status of the individual by correlating the sets of scores obtained from measures given concurrently
 - *Predictive Validity* – describes the future performance of an individual by correlating the sets of scores obtained from two measures given at a longer time interval
- **Construct Validity** – established statistically by comparing psychological traits of factors that influence scores in a test (e.g. verbal, numerical, spatial)
 - *Convergent Validity* – established if the instrument defines a similar trait other than what it intended to measure (e.g. Critical Thinking Test may be correlated with Creative Thinking Test)
 - *Divergent Validity* – established if an instrument can describe only the intended trait and no other traits (e.g. Critical Thinking Test may not be correlated with Reading Comprehension Test)

Factors Influencing the Validity of Tests

- A. Appropriateness of test

B. Directions

C. Reading vocabulary and sentence structures
Difficulty of items
- E. Construction of test items

D.F. Length of the test

G. Arrangement of items

H. Patterns of answers

Reliability

Reliability refers to the consistency of scores obtained by the same person when retested using the same instrument or one that is parallel to it.

Types of Reliability Measure

Reliability Measure	Method	Procedure
Measure of stability	Test-Retest	Give a test twice to the same group with time interval between tests from several minutes to several years
Measure of equivalence	Equivalent/Parallel Forms	Give parallel forms of tests with close time intervals between forms
Measure of stability and equivalence	Test-Retest with Equivalent Forms	Give parallel forms of tests with increased time intervals between forms
Measure of internal consistency	Split Half	Give a test once. Score equivalent halves of the test (e.g. odd- and evennumbered items)
	Kuder-Richardson	Give the test once then correlate the proportion/percentage of the students passing and not passing a given item

Factors Affecting Reliability

- A. Length of the test

B. Difficulty of the test

C. Objectivity

D. Administrability
- E. Scorability

F. Economy

G. Adequacy

H. Authenticity

Assessment in the Affective Domain

Categories of Soft Skills

Soft Skill	Observable Behavior
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Social Skills	• Provides peers with positive feedback • Offers help or assistance to peers • Initiates interactions with peers • Participates in discussion with peers • Has sense of humor, shares amusement with peers. • Has friends • Can carry out leadership activities • Engages in appropriate Social Behavior
Self- Management	• Controls displays of temper when angry • Accept legitimate rules • Compromises with others to avoid conflict • Responds in socially appropriate ways to criticism from others • Handles testing and social provocations • Cooperates with others • Maintains attention to tasks • Is respectful to teachers and staffs
Academic Soft Skills	• Works independently • Completes assigned tasks • Listens to and carries out teacher directions • Produces work of acceptable quality for ability level • Brings requires materials to school • Arrives at school on time and without due absences • Ask for assistance as needed, ask questions • Uses appropriate study skills
Approaches to Learning	• Enjoys school • Takes on challenging tasks • Has confidence in abilities • Works Hard • Is enthusiastic and adventurous • Is involved with extra- curricular activities

Affective Assessment Procedures/Tools

- **Observational Techniques**
 - Anecdotal records
 - **Peer appraisal**
 - Guess-Who technique
 - Sociometric technique
 - Self-report technique
 - Attitude scales
- **Personality Assessments**
 - Personality inventories
 - Creativity tests
 - Interest inventories

Data and Data Measures

Measures of Central Location

Any measure indicating the center of a set of data, arranged in an increasing or decreasing order of magnitude, is called a **measure of central location** or a **measure of central tendency**.

- The **arithmetic mean** is the sum of the data values divided by the total number of values.
- The **median** of a set of numbers arranged in order of magnitude is the middle value or the arithmetic mean of the two middle values.

- The **mode** is defined to be the value that occurs most often in a data set. The mode may not exist, and even if it does exist, it may not be unique.

Interpretation of Measures of Central Tendency

- The mean (or median or mode) is the point on the scale around which scores tend to group
- It is the average or typical score which represents a given group of subjects
- Given two or more values of central tendency, one can define who performed poor, good, better, or best

Measures of Variability

A **measure of variation** or **dispersion** describes how large the differences between the individual scores.

- The larger the measure of variability, the more spread the scores, and the group is said to be heterogeneous.
- The smaller the measure of variability, the less spread the scores, and the group is said to be homogenous.

Range, Standard Deviation, Quartile Deviation

- The **range** of a set of data is the difference between the largest and smallest number in the set.

- Given the finite population $X_1, X_2, \dots, X_N$, the **population standard deviation** is
$$s = \sqrt{\frac{\sum_{i=1}^N (X_i - \bar{X})^2}{N}}$$

- Quartile Deviation: $QD = \frac{Q_3 - Q_1}{2}$

Interpretation of Standard Deviation and Quartile Deviation

- Standard Deviation
 - The computed value indicates how spread the scores are. One SD below and above the mean includes around 68.26% of the cases. Measuring off two SD units on each side of the mean includes (between the two points) approximately 95% of the cases. And for three SD units, 99% of the cases are covered between the two points.
 - Helps in determining how many students performed about the mean or average, above average or below average
- Quartile Deviation
 - In a normal distribution, getting the median and adding and subtracting 1 QD on each side includes approximately 50% of the cases. In a large sample, four (4) QDs on each side of the median include practically all the cases.

Measures of Relative Position

Percentiles

- Percentiles divide the distribution into 100 groups.
- Deciles divide the data set into 10 groups. Deciles are denoted by $D_1, D_2, \dots, D_9$ with the corresponding percentiles being $P_{10}, P_{20}, \dots, P_{90}$.
- Quartiles divide the data set into 4 groups. Quartiles are denoted by Q_1, Q_2 , and Q_3 with the corresponding percentiles being P_{25}, P_{50} , and P_{75} .
- The interquartile range, $IQR = Q_3 - Q_1$.

Standard Scores

- The standard score or z-score for a value is obtained by subtracting the mean from the value and dividing the result by the standard deviation. It represents the number of standard deviations a data value falls above or below the mean.

Stanines

- Standard scores that tell the location of a raw score in a specific segment in a normal distribution which is divided into 9 segments, numbered from a low of 1 through a high of 9
- Scores falling within the boundaries of these segments are assigned one of these 9 numbers (standard nine)

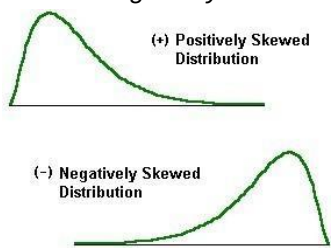
t-Score

- Tells the location of a score in a normal distribution having a mean of 50 and a standard deviation of 10

Measures of Shape

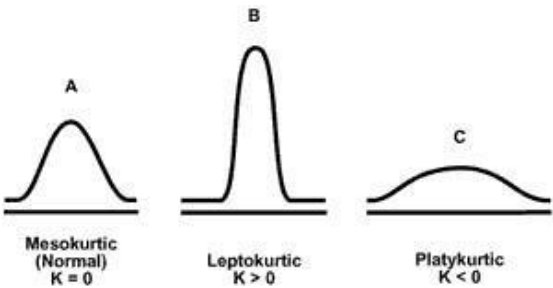
Skewness – deviation from normality in the shape of a distribution

- *Positively skewed* – most scores are low, there are extremely high scores, and the mean is greater than the mode
- *Negatively skewed* - most scores are high, there are extremely low scores, and the mean is lower than the mode



Kurtosis – the peakedness or flatness of the distribution

- *Mesokurtic* – moderate peakedness
- *Leptokurtic* – more peaked or steeper than a normal distribution
- *Platykurtic* – flatter than a normal distribution



Other Shapes

- *Bimodal* – curves with two peaks or mode
- *Polymodal* – curve with three or more modes
- *Rectangular* – there is no mode

Assigning Grades/Marks/Ratings

Marking/Grading is the process of assigning value to a performance.

Could be in	A. Percent, such as 70%, 75%, 80% B. Numbers, such as 1.0, 2.0, 3.0, 4.0 C. Letters, such as A, B, C, D or F D. Descriptive expressions, such as Outstanding (O), Very Satisfactory (VS), Satisfactory (S), Needs Improvement (NI) Note: Any symbol can be used provided that it has a uniform meaning to all concerned
Could represent	A. How a student is performing in relation to other students (Norm-referenced grading) B. The extent to which a student has mastered a particular body of knowledge (Criterion-referenced grading) C. How a student is performing in relation to a teacher's judgment of his or her potential (Grading in relations to teacher's judgment)
Could be for	A. Certification that gives assurance that a student has mastered a specific content or achieved a certain level of accomplishment B. Selection that provides basis in identifying or grouping students for certain educational paths or programs C. Direction that provides information for diagnosis and planning D. Motivation that emphasizes specific material or skills to be learned and helping students to understand and improve their performance
Could be based on	A. Examination results or test data G. Reports, themes and research papers B. Observations of student work H. Discussions and debates C. Group evaluation activities I. Portfolios D. Class discussion and recitation J. Projects E. Homework K. Attitudes F. Notebooks and note-taking

Could be assigned by using	A. Criterion-referenced grading		
	- Based on fixed or absolute standards where grade is assigned based on how a student has met the criteria or the well-defined objectives of a course that were spelled out in advance - It is then up to the student to earn the grade he/she wants to receive regardless of how other students in the class have performed.		
	B. Norm-referenced grading		
	- Based on relative standards where a student's grade reflects his/her level of achievement relative to the performance of other students in the class - In this system, the grade is assigned based on the average of test scores.		
	C. Point or Percentage grading		
	- The teacher identifies points or percentages for various tests and class activities depending on their importance. The total of these points will be the grade assigned to the student. - Example: Written Outputs (50%), Oral Outputs (30%), Special project (20%)		
	D. Contract grading		
	Each student agrees to work for a particular grade according to agreed upon standards		
	Grade	Condition	
	F	Not coming to class regularly or not turning in the required works	
	D	Coming to class regularly and turning in the required work on time	
	C	Coming to class regularly, turning in the required work on time, and receiving a check mark on all assignments to indicate they are satisfactory	
	B	Coming to class regularly, turning in the required work on time, and	
		receiving a check mark on all assignments except at least three that achieve a check-plus, indicating superior achievement	
	A	As above, plus a written report on one of the books listed for supplementary reading	

Guidelines in Grading Students

1. Explain your grading system to the students early in the course and remind them of the grading policies regularly.
2. Base grades on a predetermined and reasonable set of standards.
3. Base grades on as much objective evidence as possible.
4. Base grades on the student's relative standing compared to classmates.
5. Base grades on a variety of sources.
6. As a rule, do not change grades.
7. Become familiar with the grading policy of your school and with your colleague's standards.
8. When failing a student, closely follow school procedures.
9. Record grades on report cards and cumulative records.
10. Guard against bias in grading.
11. Keep students informed of their standing in the class.

Conducting Parent-Teacher Conferences

1. Make plans for the conference
2. Begin the conference in a positive manner
3. Present the student's strong points before describing the areas needing improvement
4. Encourage parents to participate and share information
5. Plan a course of action cooperatively
6. End the conference with a positive comment
7. Use good human relation skills during the conference
8. DOs
 - a) Be friendly and informal
 - b) Be positive in your approach
 - c) Be willing to explain in understandable terms
 - d) Be willing to listen
 - e) Be willing to accept parents' feelings
 - f) Be careful about giving advice
9. DON'Ts
 - a) Don't argue or get angry
 - b) Don't ask embarrassing questions
 - c) Don't talk about other students, teachers or parents

- d) Don't bluff if you do not know an answer
- e) Don't reject parents' suggestions
- f) Don't be a know-it-all person

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